

# Answer Key & Teacher Notes

Session 1 – Addition Equations • Session 2 – Subtraction Equations

## HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

|                                  |                                                                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Warm-Up</b>                   | First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.                                                                                                                             |
| <b>Prerequisite Practice A/B</b> | Before the lesson, for students who need the skill the session uses. Key in that folder.                                                                                           |
| <b>Guided Notes</b>              | During the lesson. Students fill in the blanks as you teach; answers below.                                                                                                        |
| <b>Practice Form A</b>           | In class, after the lesson. Two pages. The self-check code tells students if they are on track.                                                                                    |
| <b>Practice Form B</b>           | Same skills, new numbers: reteach, retake, small group or absent students.                                                                                                         |
| <b>Exit Ticket</b>               | Last 5 minutes. Two per page; cut in half.                                                                                                                                         |
| <b>Homework Packet</b>           | Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support. |

## SESSION 1 • TEACHER NOTES

Addition Equations

### TALK PROMPTS

- Why do you subtract instead of add when the variable has a number added to it?
- How does the story tell you which number is added and which number is the total?

### PARTNER TASK

Form A, problems 4 and 5. One partner solves Aquarium Fish ( $f + 14 = 62$ ); the other checks by substituting. Switch roles with Savings Jar ( $d + 16 = 73$ ).

### WATCH FOR

- Subtracting from one side only ( $b + 26 = 40$  gives  $b = 40$ ).
- Adding the number instead of subtracting it ( $y + 34 = 50$  gives  $y = 84$ ).
- Misaligning decimal points ( $5.9 - 2.4$ ).

## SESSION 2 • TEACHER NOTES

Subtraction Equations

### TALK PROMPTS

- Why does adding the same amount to both sides not change the solution?
- Why does adding back the amount used bring you to the starting amount?

### PARTNER TASK

Form A, problems 3 and 4. One partner solves Phone Storage ( $g - 12 = 29$ ); the other checks by substituting. Switch roles with Library Shelf ( $b - 38 = 54$ ).

### WATCH FOR

- Subtracting again instead of adding ( $k - 19 = 33$  gives 14).
- Accepting an answer smaller than the number on the right side of  $x - a = b$ .
- Misaligning decimal points with money ( $\$18.50 + \$31.75$ ).

## ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

| Sheet              | Standard | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|--------------------|----------|---|---|---|---|---|---|---|---|---|----|----|
| <b>S1 • Form A</b> | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 3 | 1  | 1  |
| <b>S1 • Form B</b> | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 3 | 1  | 1  |
| <b>S2 • Form A</b> | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 3 | 1  | 2  |
| <b>S2 • Form B</b> | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 3 | 1  | 1  |

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

## GUIDED NOTES

**inverse operations:** Two operations that **undo** each other.

**isolate:** Get the variable **alone** on one side.

**Subtraction Property of Equality:** Subtract the same number from **both** sides. They stay equal.

- Subtraction **undoes** addition.
- To solve  $x + 14 = 40$ , subtract **14** from both sides so the equation stays **balanced**.

**Example:** Subtract **21** from both sides.  $p = 50 - 21 = 29$ . Check:  $21 + 29 = 50$ . Answer:  $p = 29$

## YOU TRY

1.  $a. x = 16$   $b. m = 18$
2.  $k = 10 - 4.5 = 5.5$ . Check:  $5.5 + 4.5 = 10$ .

## EXIT TICKET

1.  $a. a = 27$   $b. k = 4.5$
2.  $b + 25 = 71$ ;  $b = 46$  books
3. **B** (subtract 18 from both sides)

## PRACTICE FORM A

1.  $a. n = 37$   $b. p = 25$   $c. x = 28$   $d. y = 12.5$   $e. m = 18$   $f. w = 6.5$
2.  $a. r + 19 = 56$   $b. r = 56 - 19 = 37$   $c. 37 + 19 = 56$
3.  $s + 3.5 = 12.5$ ;  $s = 9$  miles
4.  $f = 62 - 14 = 48$  fish
5.  $d = 73 - 16 = 57$  dollars
6.  $r + 2.4 = 5.9$ ;  $r = 3.5$  inches
7.  $p + 38 = 91$ ;  $p = 53$  pages. Check:  $53 + 38 = 91$ .
8.  $a. s + 46 = 128$   $b. s = 82$  dollars  $c. 128 + e = 200$ ;  $e = 72$  dollars
9. He must subtract 26 from both sides to keep the equation balanced:  $b = 40 - 26 = 14$ . Check:  $14 + 26 = 40$ .
10. **B** ( $y = 27$ )
11. **C** (subtract the same number from both sides)

**Self-check word** **BALANCE**

## PRACTICE FORM B

1.  $a. n = 37$   $b. p = 27$   $c. x = 33$   $d. y = 11.5$   $e. m = 26$   $f. w = 7.5$
2.  $a. r + 23 = 61$   $b. r = 61 - 23 = 38$   $c. 38 + 23 = 61$
3.  $s + 4.5 = 14.5$ ;  $s = 10$  miles
4.  $f = 63 - 17 = 46$  fish
5.  $d = 81 - 18 = 63$  dollars
6.  $r + 1.8 = 6.3$ ;  $r = 4.5$  inches
7.  $p + 42 = 97$ ;  $p = 55$  pages. Check:  $55 + 42 = 97$ .
8.  $a. s + 54 = 135$   $b. s = 81$  dollars  $c. 135 + e = 200$ ;  $e = 65$  dollars
9. Adding 34 does not undo  $+ 34$ . He should subtract 34 from both sides:  $y = 50 - 34 = 16$ . Check:  $16 + 34 = 50$ .
10. **B** ( $b = 15$ )
11. **C** (balanced)

**Self-check word** **INVERSE**

## GUIDED NOTES

**Addition Property of Equality:** Add the same number to **both** sides. They stay equal.

**difference:** The answer when you **subtract**.

**isolate:** Get the variable **alone** on one side.

- Addition **undoes** subtraction.
- To solve  $p - 7 = 12$ , add **7** to both sides.

**Example:** Add **9** to both sides.  $m = 14 + 9 = 23$ . Check:  $23 - 9 = 14$ . Answer:  $m = 23$

## YOU TRY

1.  $a \cdot x = 26$  b.  $n = 55$
2.  $p = 4 + 2.5 = 6.5$ . Check:  $6.5 - 2.5 = 4$ .

## EXIT TICKET

1.  $a \cdot n = 42$  b.  $p = 9.5$
2.  $s - 87 = 26$ ;  $s = 113$  snacks
3. **B** (add 19 to both sides)

## PRACTICE FORM A

1.  $a \cdot n = 47$  b.  $p = 14$  c.  $k = 65$  d.  $m = 45$  e.  $r = 10$  f.  $y = 75$
2.  $a \cdot p - 47 = 16$  b.  $p = 16 + 47 = 63$  c.  $63 - 47 = 16$
3.  $g - 12 = 29$ ;  $g = 41$  gigabytes
4.  $b = 54 + 38 = 92$  books
5.  $n = 89 + 145 = 234$  miles
6.  $f - 18.50 = 31.75$ ;  $f = \$50.25$
7.  $a - 128 = 95$ ;  $a = 223$  pounds. Check:  $223 - 128 = 95$ .
8.  $a \cdot b - 34 = 89$  b.  $b = 123$  books c.  $89 - 45 = r$ ;  $r = 44$  books
9. Subtracting 19 does not undo  $- 19$ . She should add 19 to both sides:  $k = 33 + 19 = 52$ . Check:  $52 - 19 = 33$ .
10. **B** ( $m = 55$ )
11. **C** (greater than 30)

**Self-check word ISOLATE**

## HOMEWORK PACKET • ANSWERS

*same answers for the core packet and Level 1*

## SHEET 1 • ADDITION EQUATIONS

1.  $a \cdot x = 8$  b.  $n = 18$  c.  $y = 35$  d.  $m = 37$  e.  $k = 5.5$  f.  $w = 4.25$
2.  $p + 46 = 93$ ;  $p = 47$
3.  $b + 27 = 64$ ;  $b = 37$  books. Check:  $37 + 27 = 64$ .
4.  $38.50 + m = 52$ ;  $m = \$13.50$ . Check:  $38.50 + 13.50 = 52$ .
5. No.  $n = 32$  is correct. Dante checked with the wrong operation. He should substitute 32 into the original equation:  $32 + 18 = 50$ , which is true.

## PRACTICE FORM B

1.  $a \cdot n = 43$  b.  $p = 11$  c.  $k = 81$  d.  $m = 55$  e.  $r = 7$  f.  $y = 85$
2.  $a \cdot p - 38 = 24$  b.  $p = 24 + 38 = 62$  c.  $62 - 38 = 24$
3.  $g - 15 = 34$ ;  $g = 49$  gigabytes
4.  $b = 57 + 42 = 99$  books
5.  $n = 75 + 160 = 235$  miles
6.  $f - 24.25 = 30.50$ ;  $f = \$54.75$
7.  $a - 136 = 89$ ;  $a = 225$  pounds. Check:  $225 - 136 = 89$ .
8.  $a \cdot b - 29 = 96$  b.  $b = 125$  books c.  $96 - 40 = r$ ;  $r = 56$  books
9. Subtracting 24 does not undo  $- 24$ . He should add 24 to both sides:  $x = 60 + 24 = 84$ . Check:  $84 - 24 = 60$ .
10. **B** ( $x = 26$ )
11. **A** (add the same number to both sides)

**Self-check word SOLVING**

## SHEET 2 • SUBTRACTION EQUATIONS

1.  $a \cdot n = 45$  b.  $k = 85$  c.  $p = 9.5$
2.  $t - 54 = 35$  (or  $t - 35 = 54$ );  $t = 89$
3.  $g - 18.25 = 31.75$ ;  $g = \$50.00$
4.  $c - 26 = 59$ ;  $c = 85$  cars. Check:  $85 - 26 = 59$ .
5. Leo.  $84 - 24 = 60$  is true. Mia subtracted instead of adding:  $36 - 24 = 12$ , not 60.
6. **C** ( $p = 74$ )
7. **B** ( $m - 48 = 37$ )